Bioinformatic approaches to regulatory genomics and epigenomics

376-1347-00L | week 04

Pierre-Luc Germain

ETH Zürich

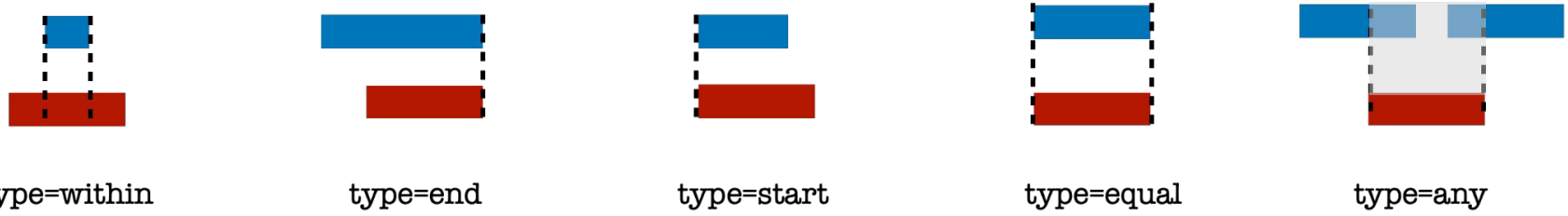
Plan

- Debriefing on the assignment
- Recap on bigwigs and visualization
- The ‘histone code’ & functional elements
- Comparing regions and signals

Debriefing on the assignments: findOverlaps

- Overlap functions (e.g. `findOverlaps`, `overlapsAny`, etc.) have a `type` argument

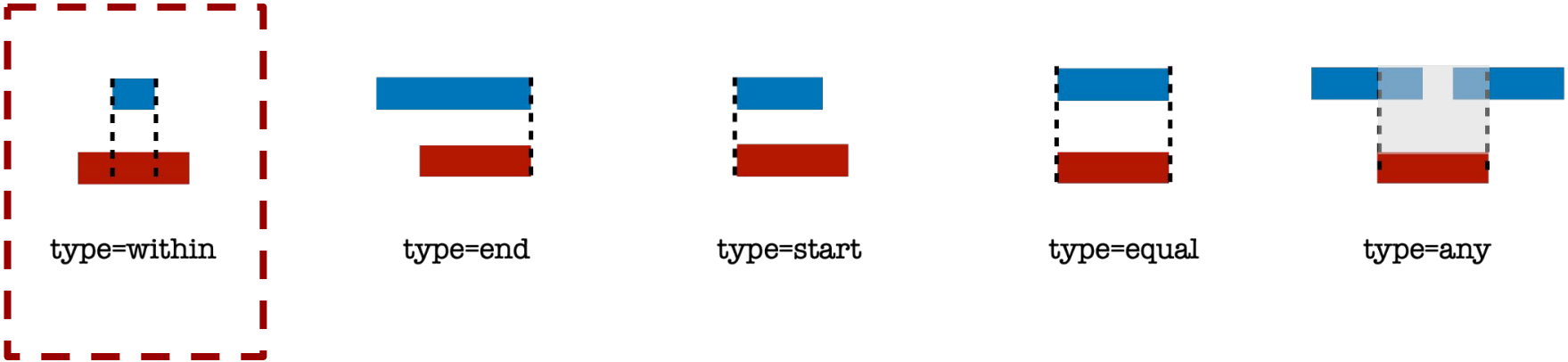
```
findOverlaps(peaks, genes, type="<...>")
```



Debriefing on the assignments: findOverlaps

- Plot the signal around one of the peaks that is located **inside** a gene.

```
findOverlaps(peaks, genes, type="<...>")
```



Practical part 1:
Recap on bigwig files and visualization

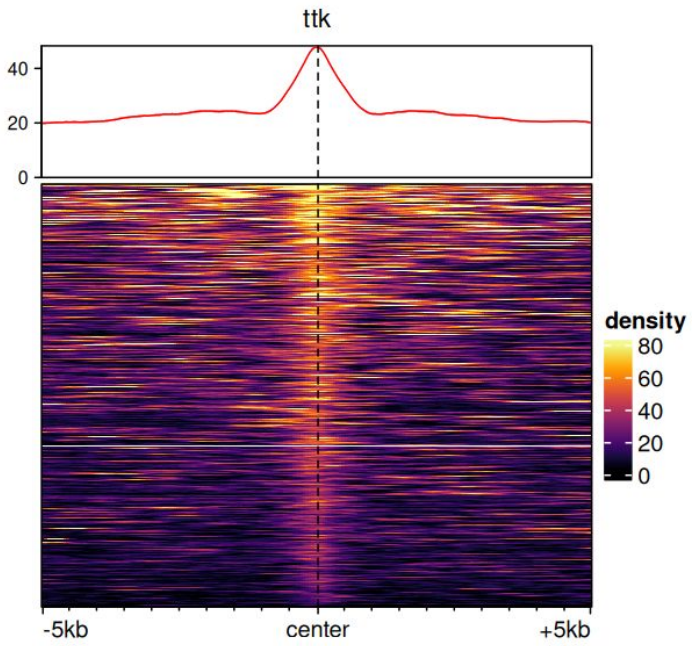
Extension of single-end reads in coverage track generation

Coverage without read extension

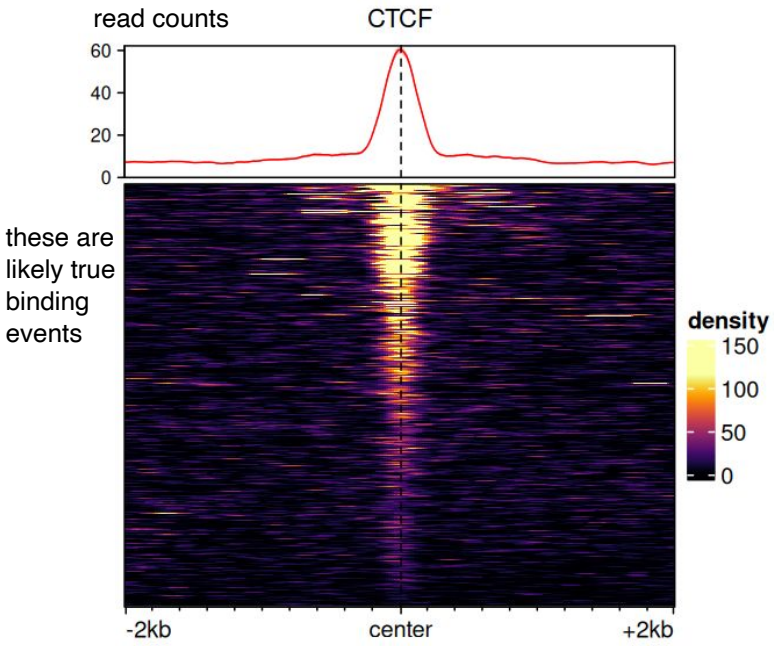
Coverage with read extension

Ttk ChIP from last week:
low quality enrichment
(poor localization of the signal
in the peak regions)

CTCF ChIP from this week:
high quality enrichment



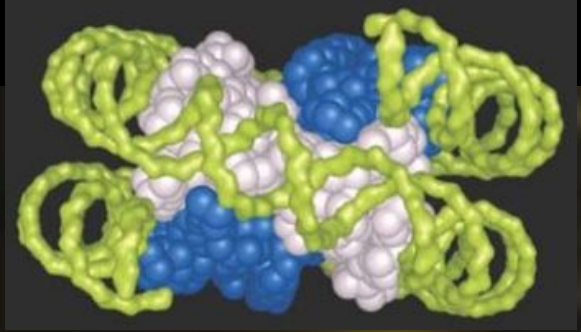
this is very likely just noisy signal

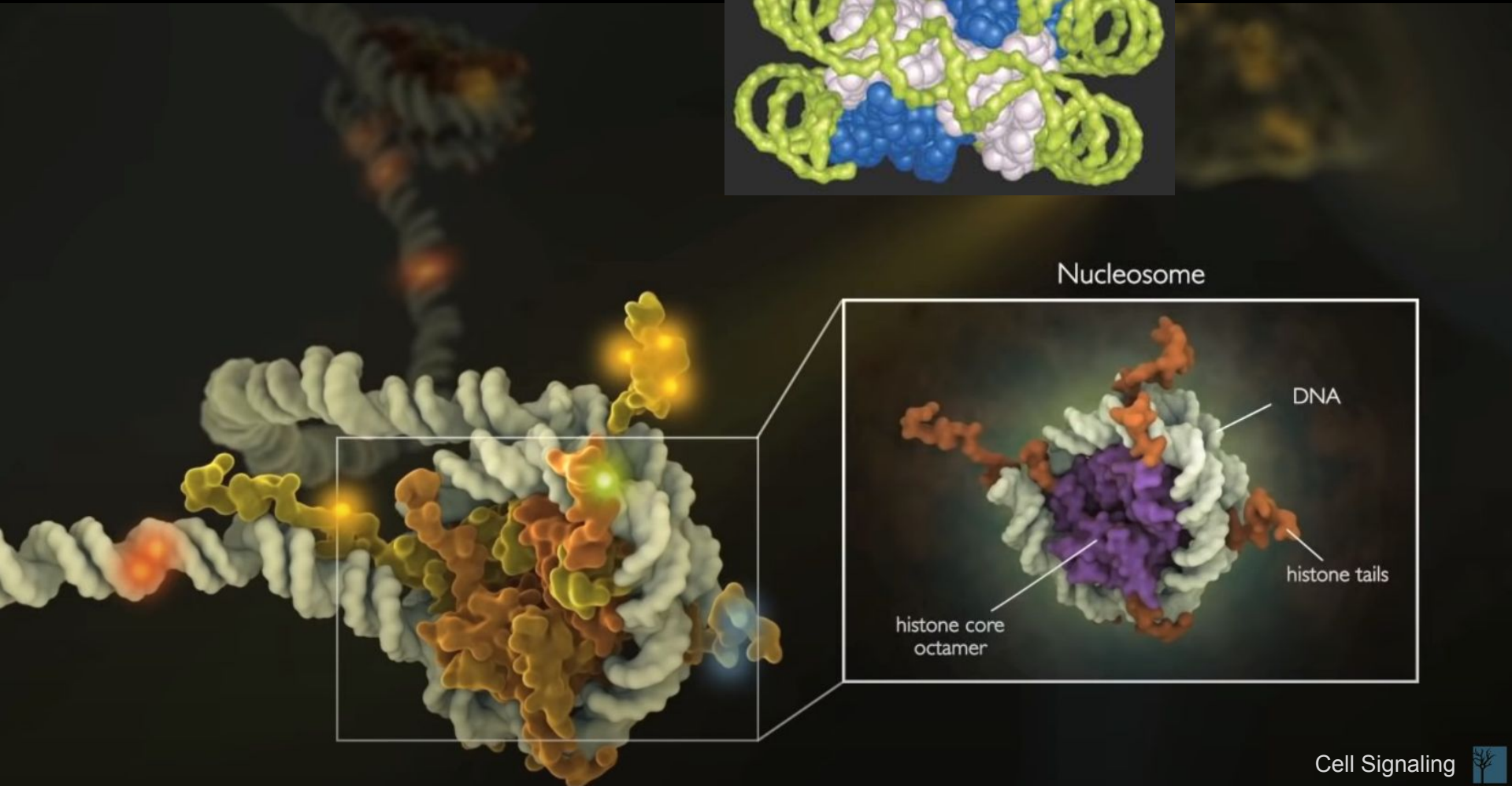


The histone code and functional elements

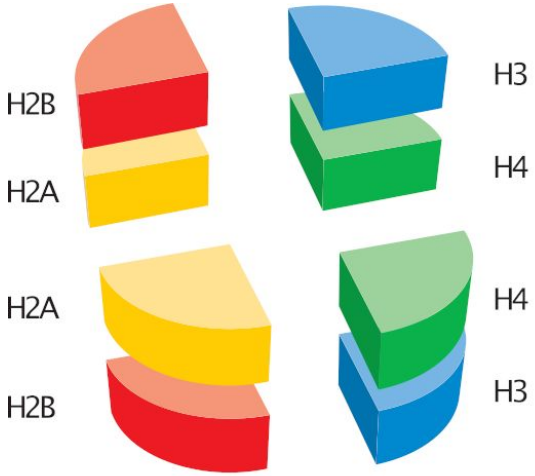
Nucleosome

(Krebs et al., 2018)

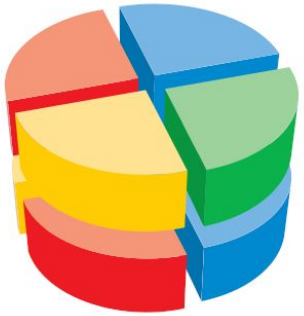




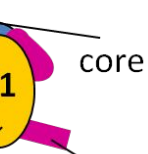
Cell Signaling



core histones



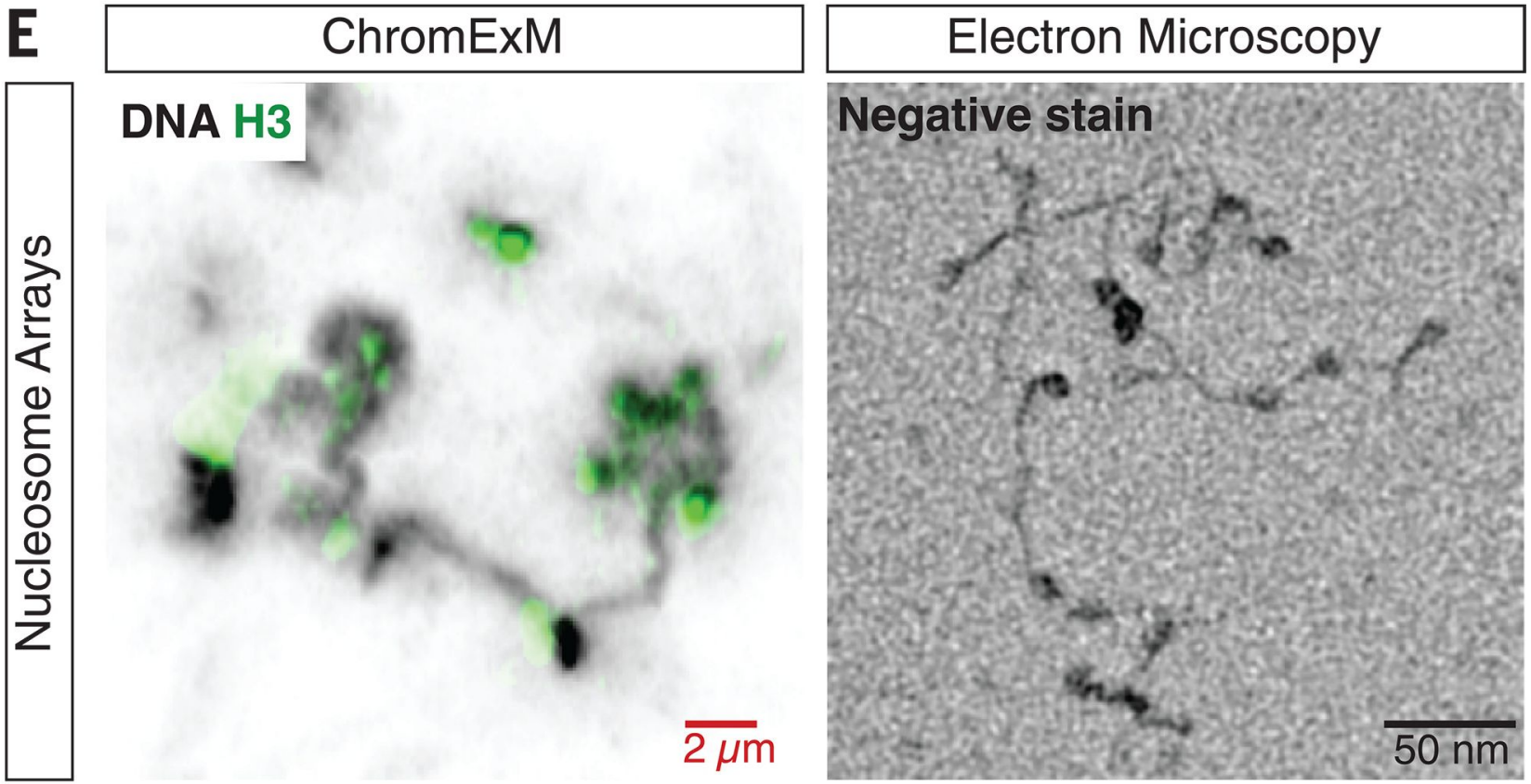
histone octamer



beads–on–a–string nucleosome array

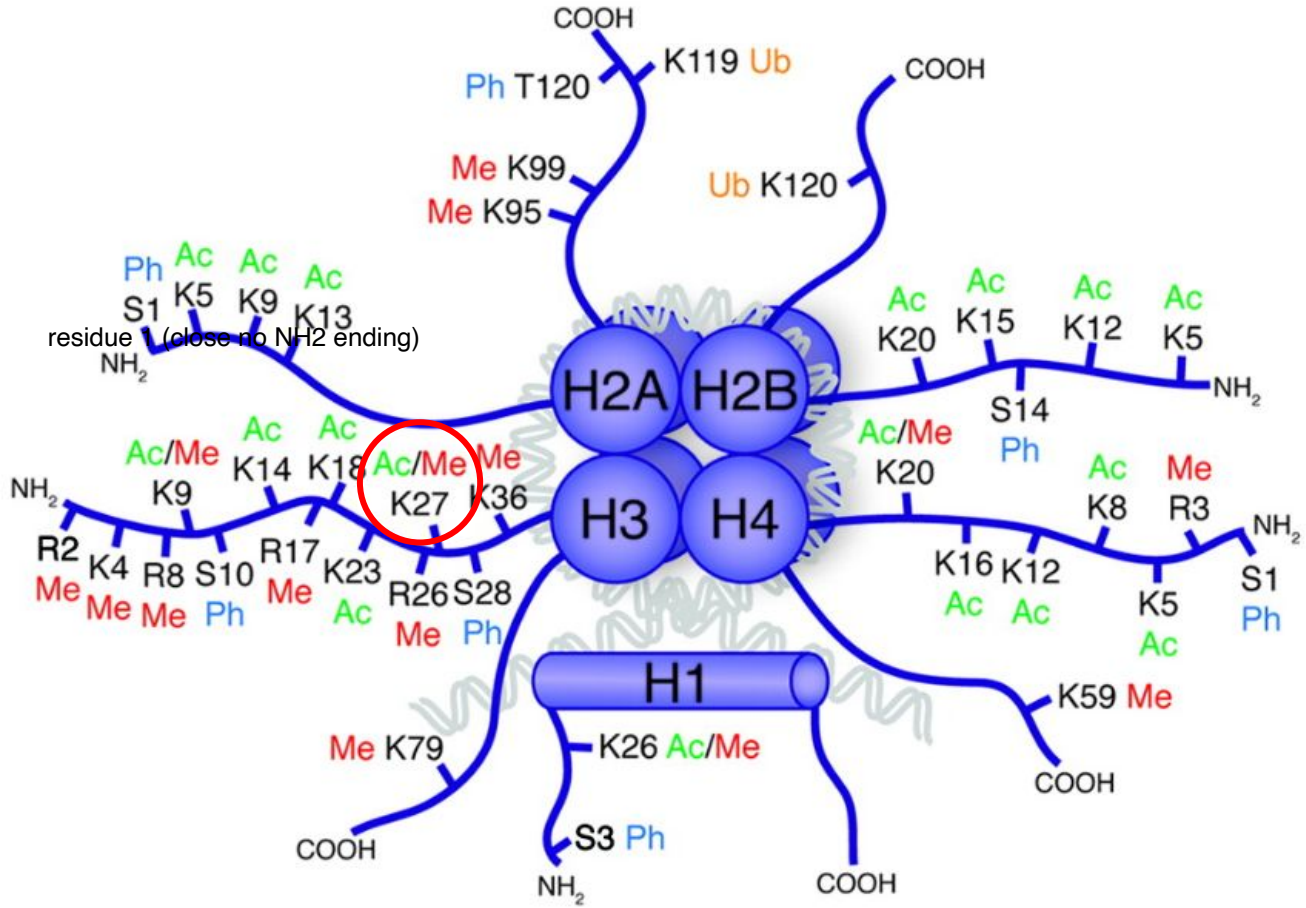
(Adapted from David O Morgan - The Cell Cycle. Principles of Control. Wikimedia Commons)

Viewing the beads-on-a-string of DNA and nucleosomes



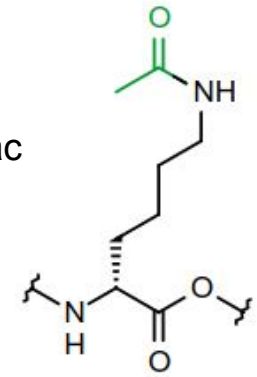
(Adapted from Pownall et al., Science 2023)

Many residues on the histone tails can be post-translationally modified

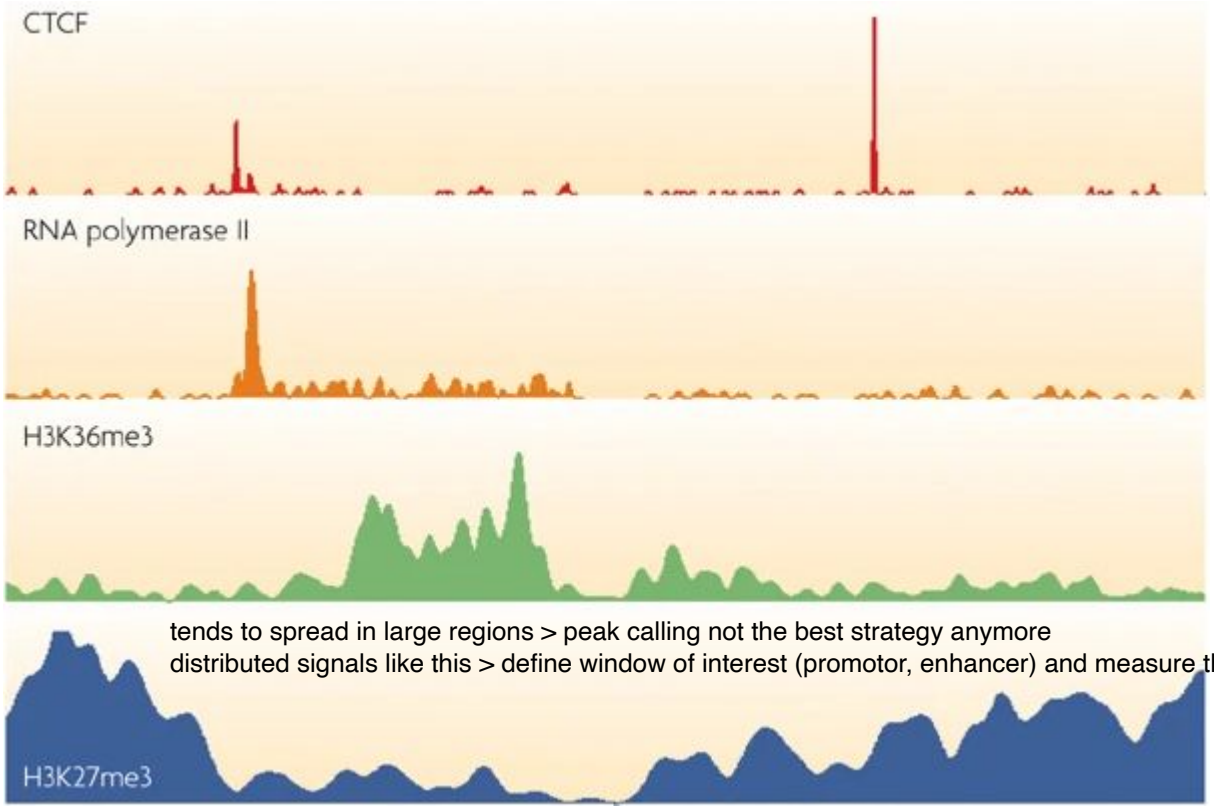


(Adapted from Tollervey & Lunyak, Epigenetics 2012)

H3K27ac



Some histone modifications appear to be very localized, e.g. happening on a specific nucleosome, while most are much more broadly distributed

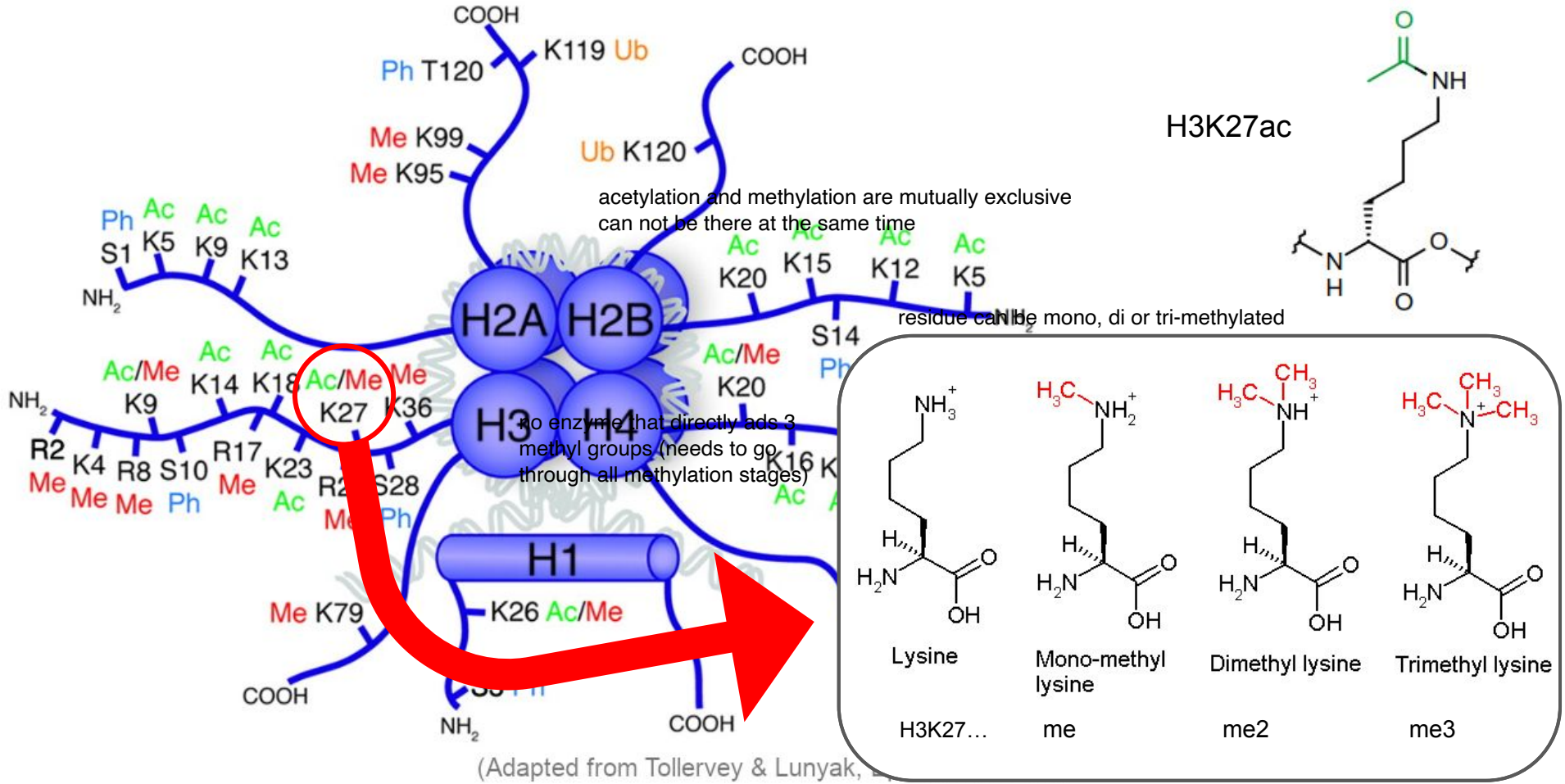


tends to spread in large regions > peak calling not the best strategy anymore
distributed signals like this > define window of interest (promotor, enhancer) and measure the signal there

The strategy of calling ‘peaks’ must therefore be adapted
(e.g. “broad” option of most peak-callers)

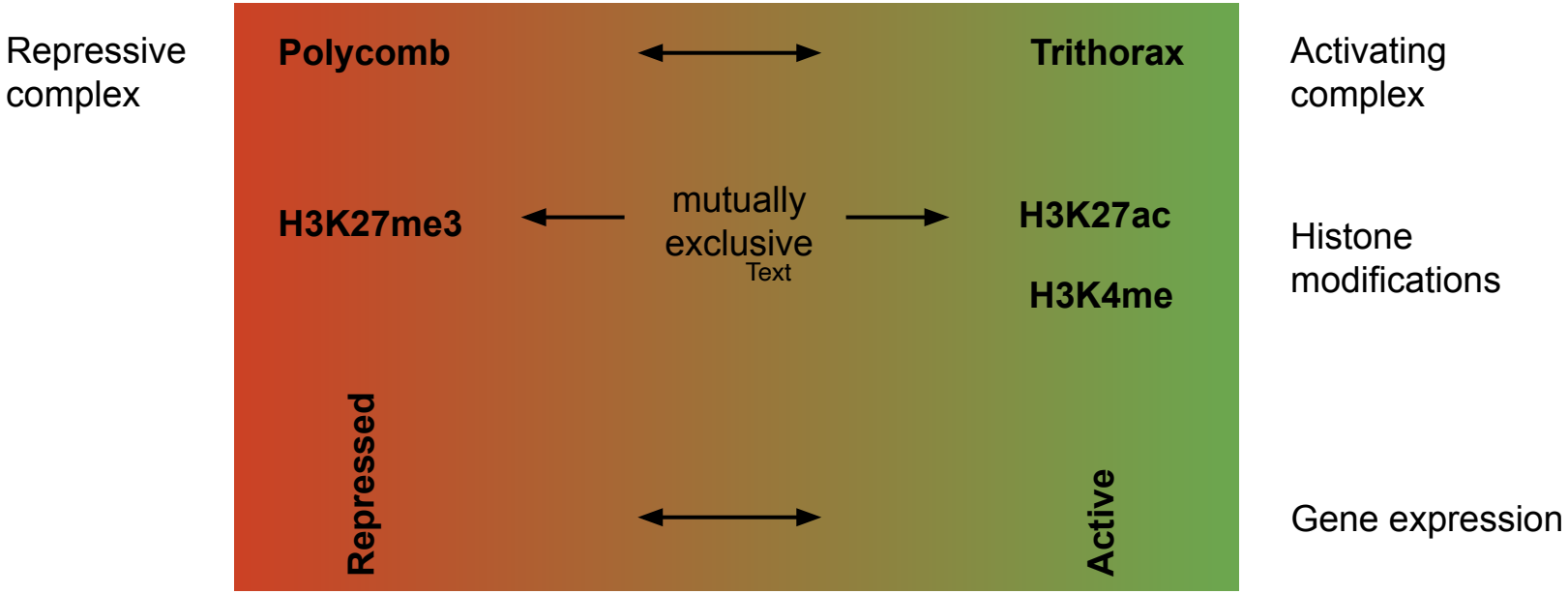
(Park, Nat Rev Gen 2009)

Many residues on the histone tails can be post-translationally modified



(Adapted from Tollervey & Lunyak,

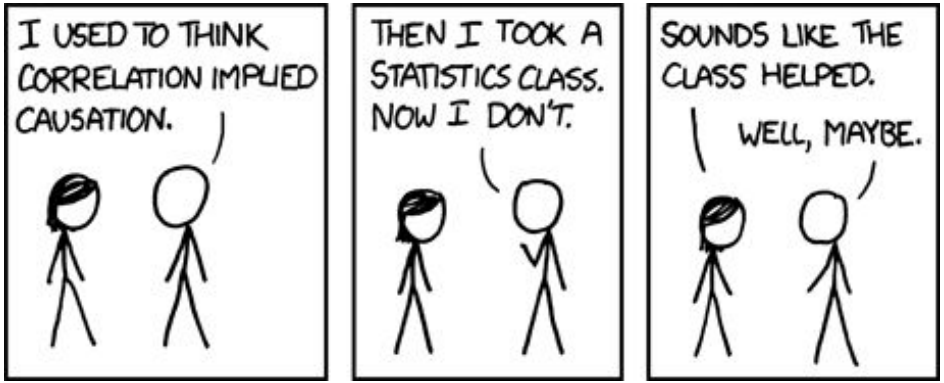
There is a very strong association of certain histone marks and activation or repression



But which comes first?

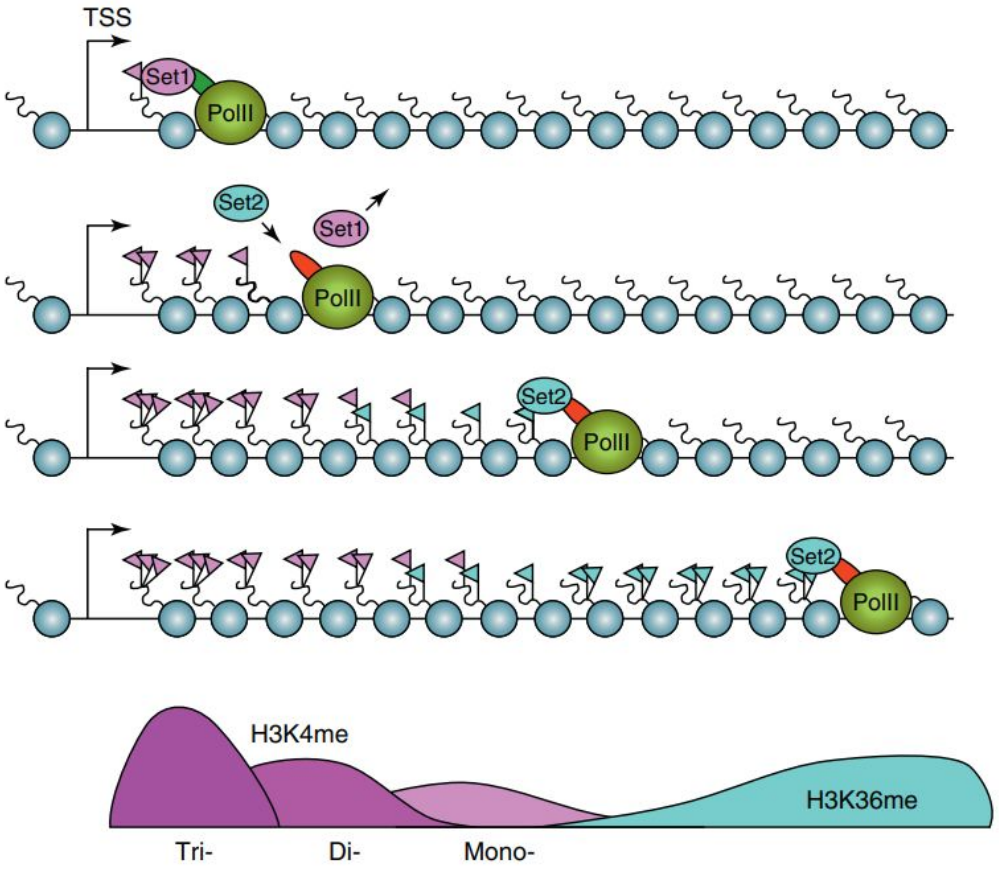
Causality or correlation?

Are histone modifications **responsible** for activation/repression, or are they merely associated **side-effects**?



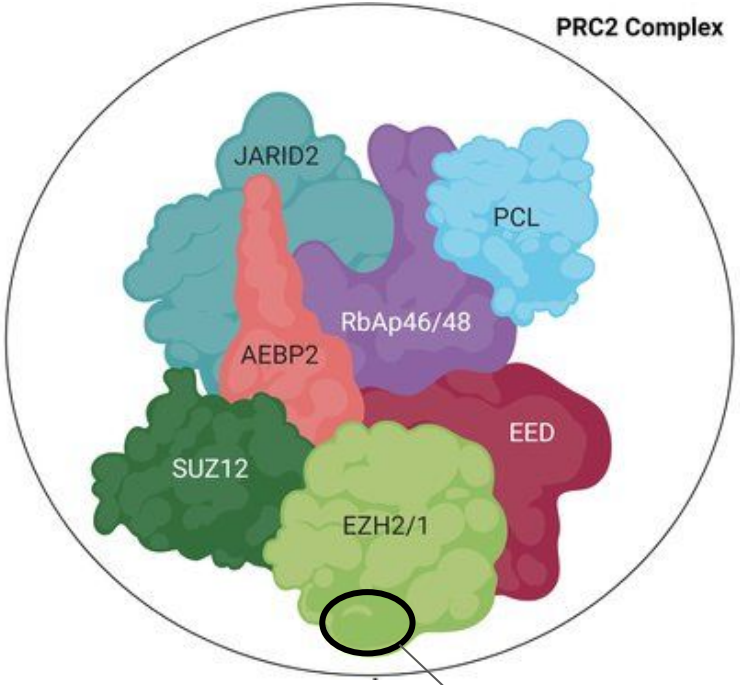
(https://xkcd.com/552)

Transcription-mediated histone modification



(Henikoff and Shilatifard 2011)

The example of H3K27me3, chiefly deposited by the polycomb repressive complex (PRC2)



Abolishing the enzymatic activity of *Ezh2*, the gene responsible for depositing H3K27me3, abolishes (most of) the mark but does not prevent the repression of the target genes, nor cellular reprogramming

(Fragola et al., PLoS Genetics 2013)

EHZ2's SET domain catalyzes the addition of a 3rd methyl group to H3K27, i.e. H3K27me2 → H3K27me3

(Adapted from Zoroddu et al., 2021)

Causality or correlation?

Other examples…

For H3K27ac, blocking the deposition of the mark does not prevent pioneering factors (PF) from opening local chromatin, whereas blocking the factors prevents H3K27ac to most sites

(Miao et al., Molecular Cell 2022)

Similarly, the loss of H3K4me3 appears to have no effect on nascent transcription

(Murray et al., bioRxiv 2019)

nature

Article | Open Access | Published: 01 March 2023

H3K4me3 regulates RNA polymerase II promoter-proximal pause-release

Hua Wang, Zheng Fan, Pavel V. Shliaha, Matthew Miele, Ronald C. Hendrickson, Xuejun Jiang & Kristian Helin

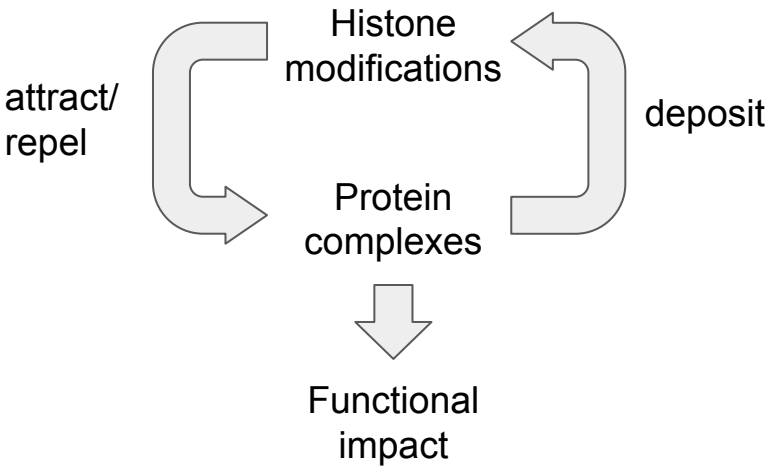
Nature **615**, 339–348 (2023)

"acute **loss of H3K4me3 does not have detectable effects on transcriptional initiation** but leads to a widespread decrease in transcriptional output, an increase in RNA polymerase II (RNAPII) pausing and slower elongation. We show that H3K4me3 is required for the recruitment of the integrator complex subunit 11 (INTS11), which is **essential for the eviction of paused RNAPII and transcriptional elongation**."

Causality or correlation?

Most likely somewhere in the middle, depending on the modification/context

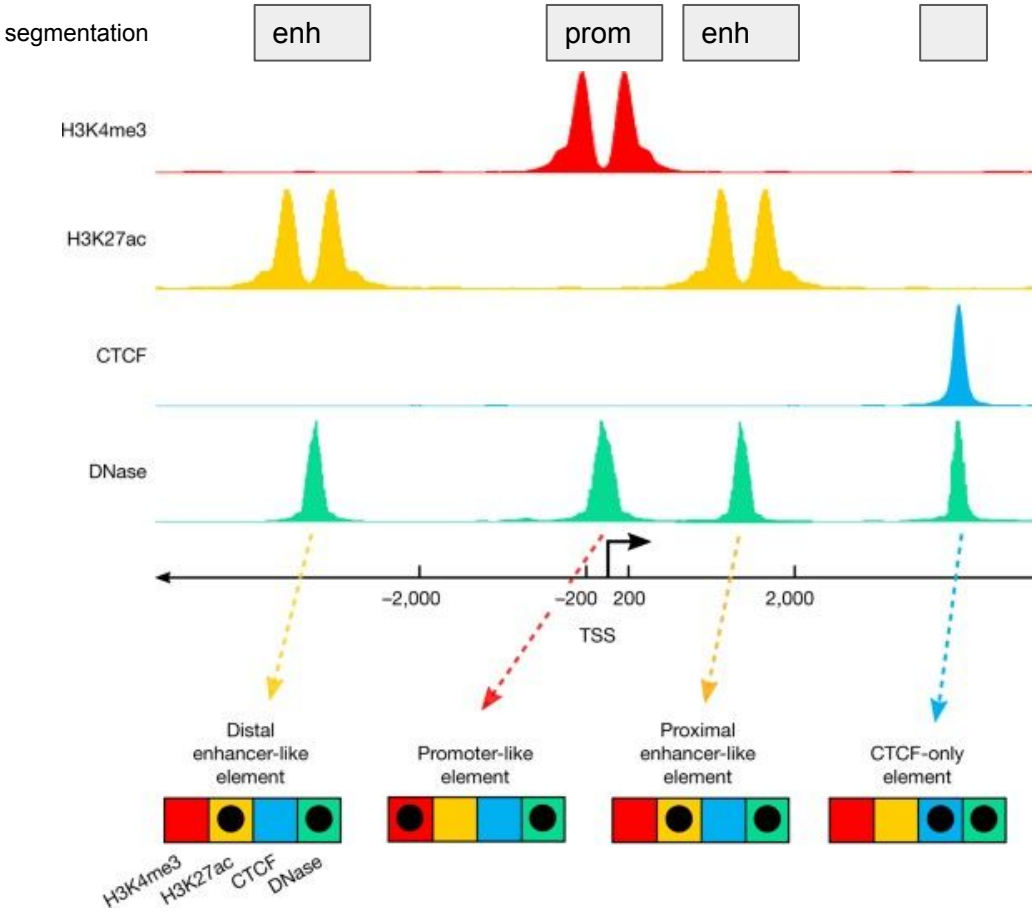
facilitate the assembly of protein complexes
but the protein complexes are actually the ones that have an effect
so the histone modification act as „lubricant“



Whether they're causative or not, they can serve as **proxies** for function.

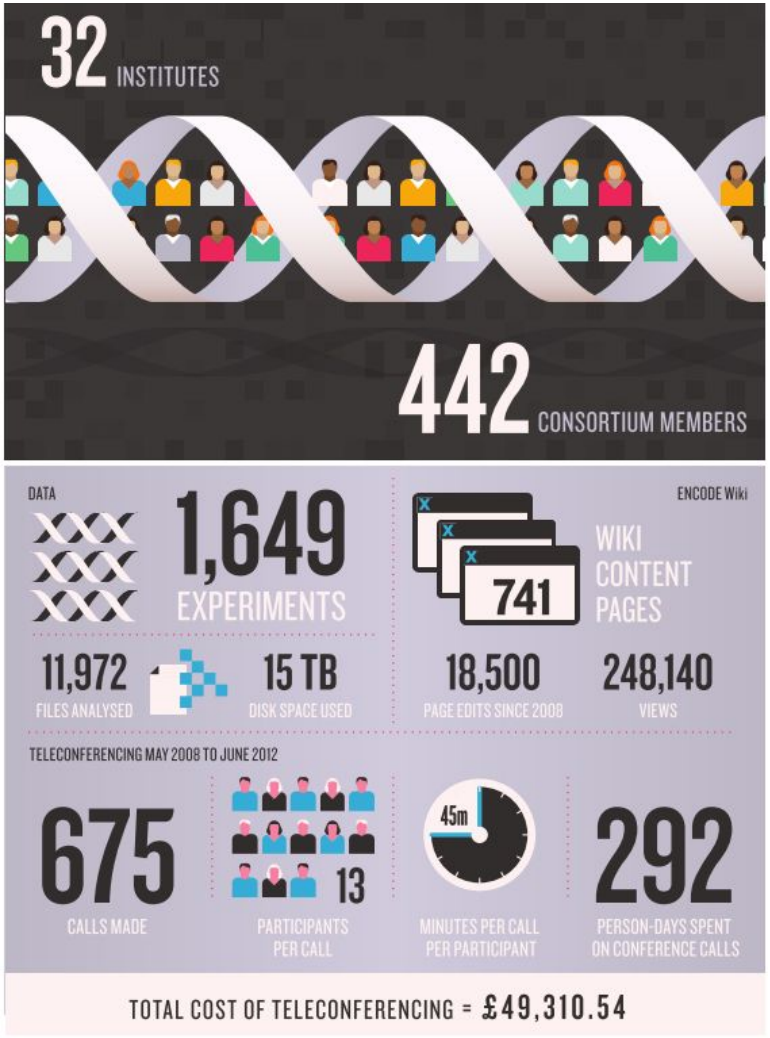
This means that profiling a few histone modifications gives an overview of the epigenomic landscape of a cellular state which would otherwise require profiling all the potentially-relevant factors/complexes

A signature-based encyclopedia of DNA elements



The “signature strategy”:

- Different types of functional genetic elements are associated with different chemical signatures
- We can identify functional elements by identifying these signatures genome-wide



The ENCyclopedia Of DNA Elements

~30 publications in September 2012

$288 million USD

… then an ENCODE2, 3, now working towards the 5…

An integrated encyclopedia of DNA elements in the human genome

The ENCODE Project Consortium

Nature **489**, 57–74 (2012) | Cite this article

Bits of Mystery DNA, Far From 'Junk,' Play Crucial Role

The New York Times

by Gina Kolata

"At least 80 percent of this DNA is *active* and *needed*."

The evolutionary arguments for junk:

- 1% protein-coding
- ~4 to 10% evolutionarily conserved
- >50% transposable elements
- Onions have a 5 times bigger genome

The very angry response:

- Graur et al., GBE 2013

NEWS&ANALYSIS

GENOMICS

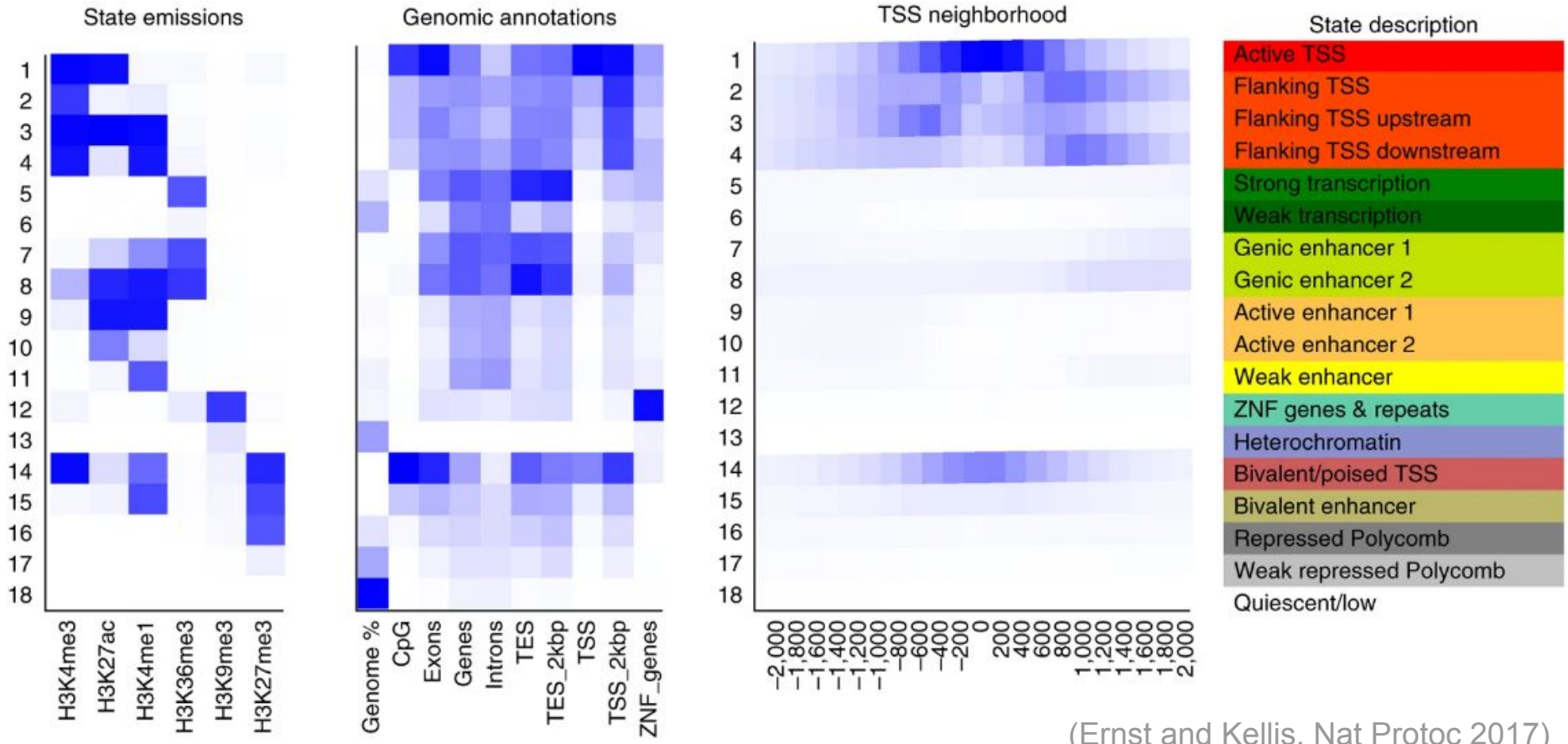
ENCODE Project Writes Eulogy For Junk DNA

–ELIZABETH PENNISI

SCIENCE VOL 337 7 SEPTEMBER 2012

(For more on the topic, see Germain, Ratti and Boem 2014; Ratti and Germain 2022)

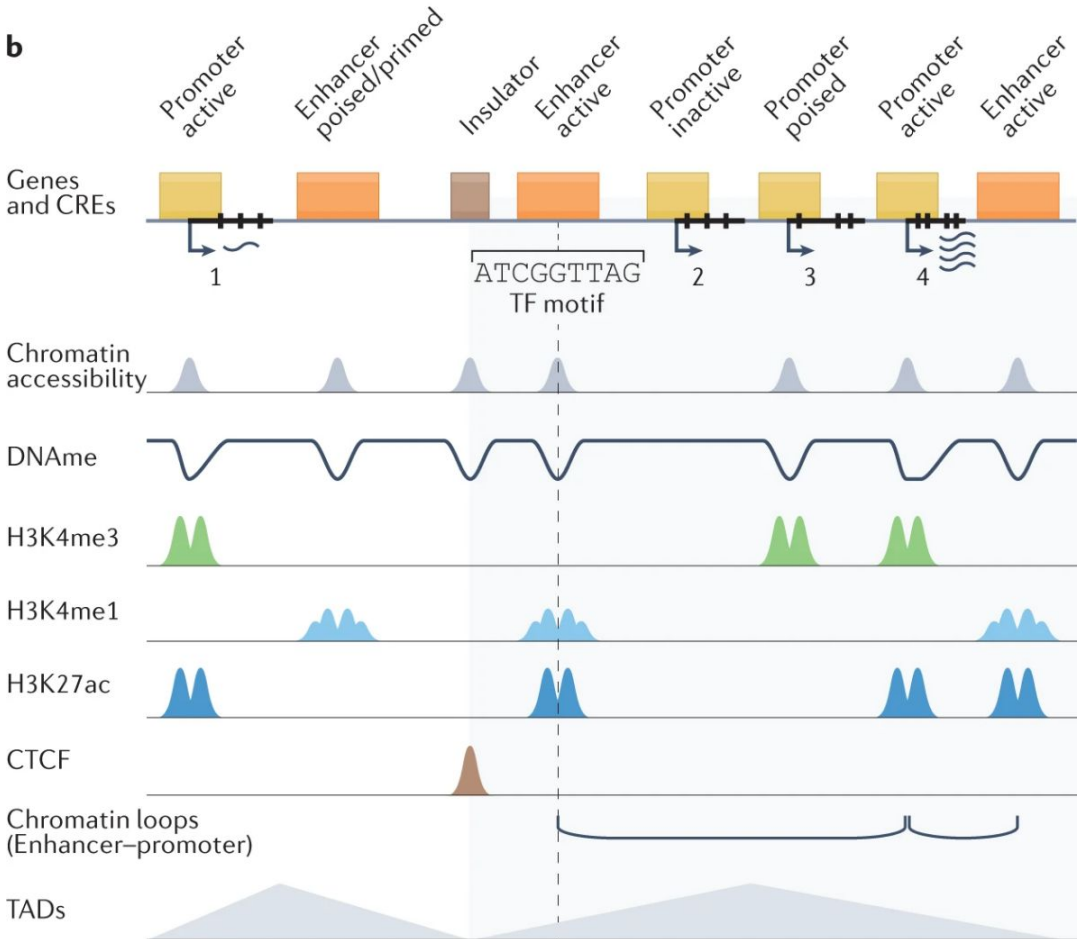
So how many kinds of functional elements/states are there?



(Ernst and Kellis, Nat Protoc 2017)

So how many kinds of functional elements/states are there?

More recent papers/reviews tend to narrow down to a very basic set of elements…



(Preissl, Gaulton & Ren, Nat Rev Gen 2023)

Some stuff is very clear:

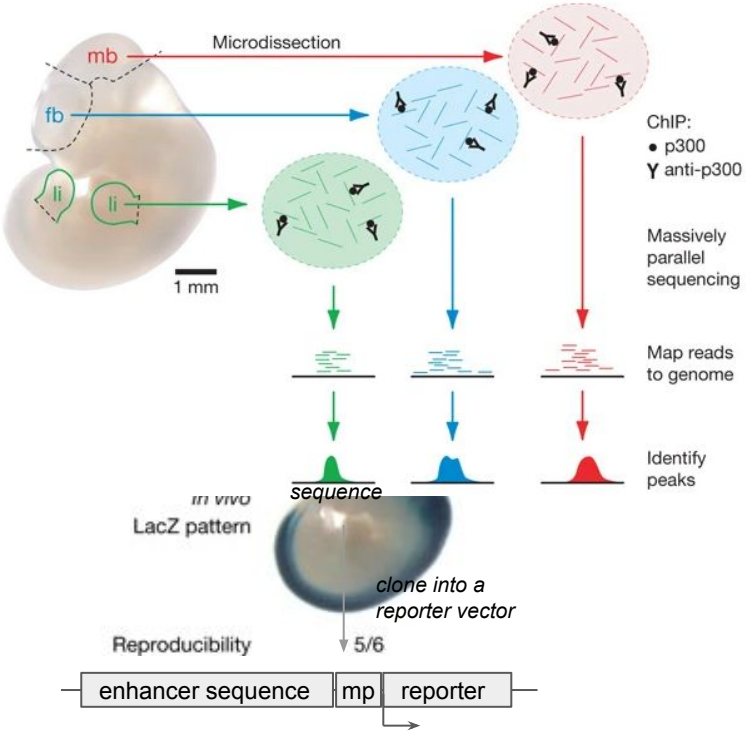
- **Transcription start site (TSS):**
  - **H3K4me3** is almost always associated with active/poised TSS
  - Active TSS are marked by **H3K27ac**
  - So-called “poised” (or bivalent) TSS are instead marked by both **H3K4me3** and **H3K27me3**

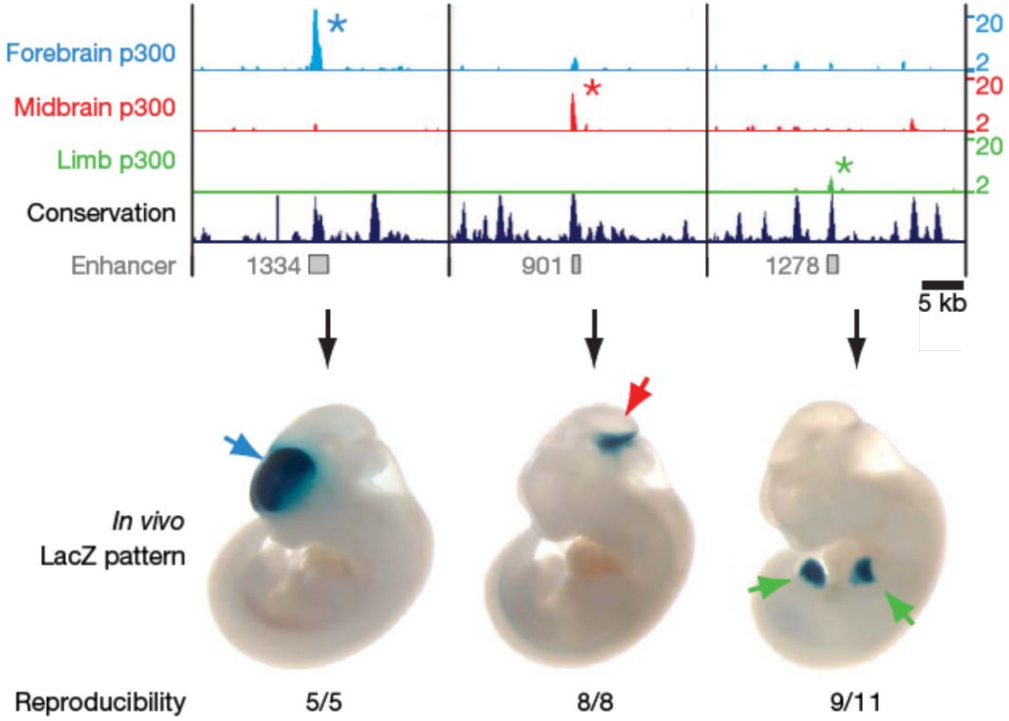
- **Enhancers:**
  - Most enhancers have **H3K4me1**
  - Active enhancers are marked by **H3K27ac**
  - So-called “poised” (or bivalent) enhancers are marked by **H3K4me1** and **H3K27me3**

- Repressed elements are marked by **H3K27me3**

- Heterochromatin is marked by **H3K9me3**
- Insulators: CTCF (+cohesin etc.)

p300 and validation of enhancer activity





(Adapted from Visel et al., 2009)

Vista enhancers: http://enhancer.lbl.gov

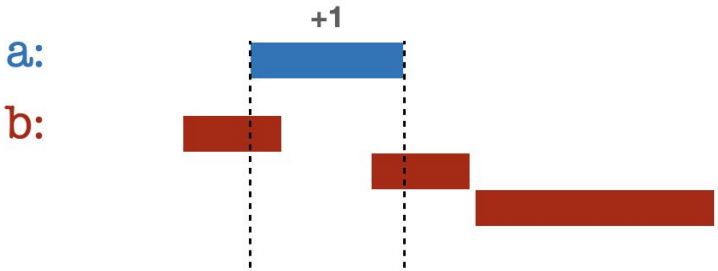
Practical part 2:
Comparing different signals

The overlap of ranges is not symmetrical:

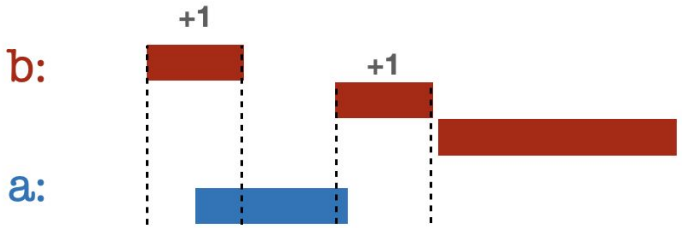
Symmetry of overlaps:

a: query
b: subject

overlapsAny(a, b)
findoverlaps(a, b)



sum(overlapsAny(a, b))=1



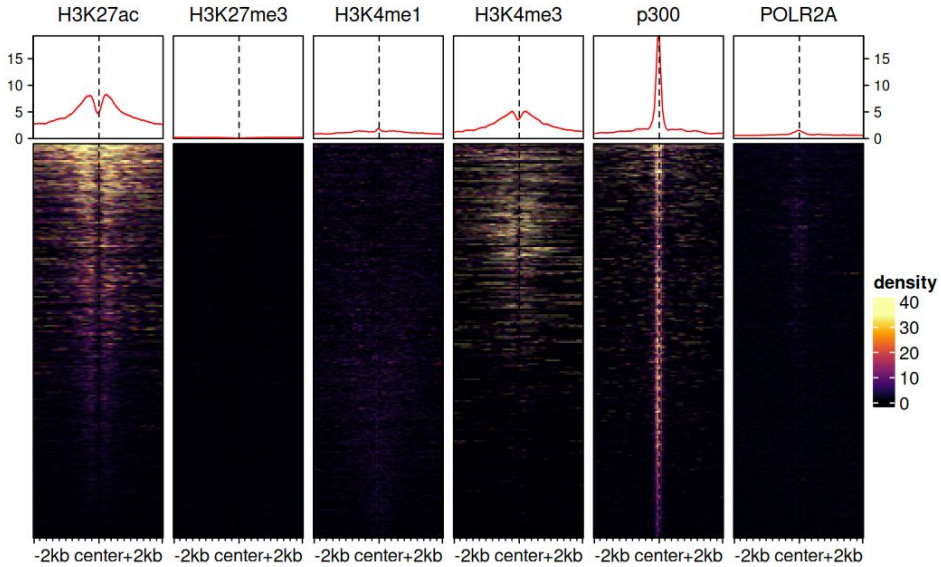
sum(overlapsAny(b, a))=2

Poor heatmap rasterization issue

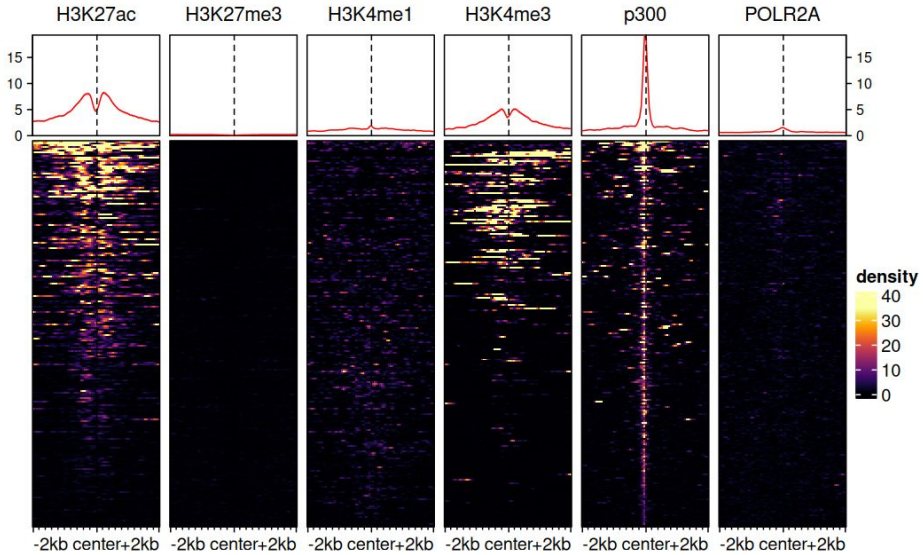
If you get this message, make sure you have the magick package installed:

```
'magick' package is suggested to install to give better rasterization.
```

With the ‘magick’ package installed :



Without :



Finding data

Generic repositories for NGS data

https://www.ebi.ac.uk/biostudies/arrayexpress

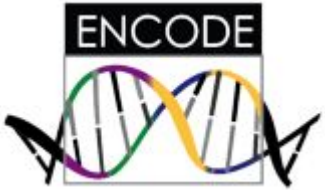
https://www.ncbi.nlm.nih.gov/geo/

https://www.ebi.ac.uk/ena/

https://www.ncbi.nlm.nih.gov/sra

International Nucleotide Sequence Database Collaboration

Quality-controlled and uniformly processed human and mouse NGS datasets



www.roadmapepigenomics.org

www.encodeproject.org

(hematopoietic system)

Assignment

- Find and download from ENCODE the **peaks** (i.e. bed-like format) for **p300**, **H3K27ac** and **H3K27me3** in mouse embryonic stem cells (mESC) and in one differentiated tissue (heart or liver):
  - (when there are replicates, we recommend using the bed file denoted as "`conservative IDR thresholded peaks`")
- Compare the signals across the two tissues, answering the following questions:
  - What is the proportion of TSS that overlap H3K27ac, H3K27me3, and the combination of both, in each tissue?
  - What is the relationship between the 3 signals (p300, H3K27ac, and H3K27me3), and is it the same in both tissues?
  - What is the relationship between regulatory elements active in mESC and in the differentiated tissue?
- For each question, provide figures and/or numbers (e.g. proportions of overlap, etc.), and write a paragraph describing your observations and what you think they mean.
- Don't forget to upload your assignment as "`assignment.html`" !